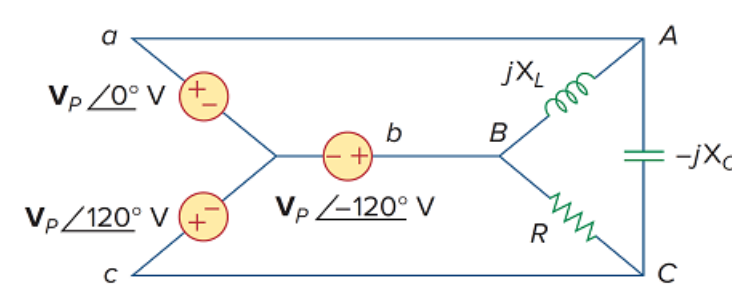


Problem

Using Fig. 12.63, design a problem to help other students to better understand unbalanced three-phase systems.

Figure 12.63



Step-by-step solution

Step 1 of 5

Using Fig. 12.63, design a problem to help other students to better understand unbalanced three-phase systems.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Refer to the unbalanced circuit of Fig. 12.63. Calculate:

(a) the line currents

Step 2 of 5

(b) the real power absorbed by the load

Step 3 of 5

(c) the total complex power supplied by the source

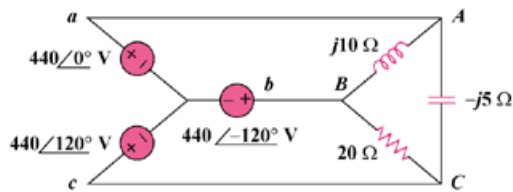
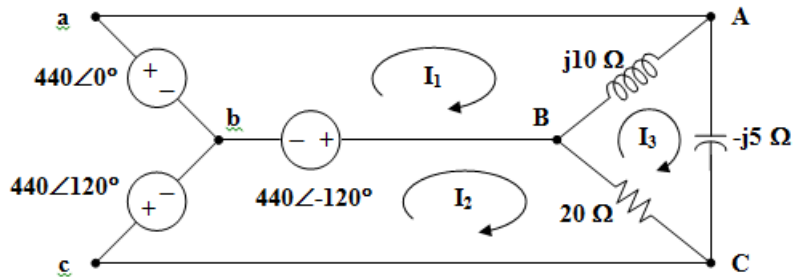


Figure 12.63

Solution

(a) Consider the circuit below.



For mesh 1,

$$440\angle -120^\circ - 440\angle 0^\circ + j10(I_1 - I_3) = 0$$

$$I_1 - I_3 = \frac{(440)(1.5 + j0.866)}{j10} = 76.21\angle -60^\circ \quad (1)$$

For mesh 2,

$$440\angle 120^\circ - 440\angle -120^\circ + 20(I_2 - I_3) = 0$$

$$I_3 - I_2 = \frac{(440)(j1.732)}{20} = j38.1 \quad (2)$$

For mesh 3,

$$j10(I_3 - I_1) + 20(I_3 - I_2) - j5I_3 = 0$$

Substituting (1) and (2) into the equation for mesh 3 gives,

$$I_3 = \frac{(440)(-1.5 + j0.866)}{j5} = 152.42\angle 60^\circ \quad (3)$$

From (1),

$$I_1 = I_3 + 76.21\angle -60^\circ = 114.315 + j66 = 132\angle 30^\circ$$

From (2),

$$I_2 = I_3 - j38.1 = 76.21 + j93.9 = 120.93\angle 50.94^\circ$$

$$I_a = I_1 = 132\angle 30^\circ \text{ A}$$

$$I_b = I_2 - I_1 = -38.105 + j27.9 = 47.23\angle 143.8^\circ \text{ A}$$

$$I_c = -I_2 = 120.9\angle 230.9^\circ \text{ A}$$

Step 4 of 5

$$(b) \quad S_{AB} = |I_1 - I_3|^2 (j10) = j58.08 \text{ kVA}$$

$$S_{BC} = |I_2 - I_3|^2 (20) = 29.04 \text{ kVA}$$

$$S_{CA} = |I_3|^2 (-j5) = (152.42)^2 (-j5) = -j116.16 \text{ kVA}$$

$$S = S_{AB} + S_{BC} + S_{CA} = 29.04 - j58.08 \text{ kVA}$$

$$\text{Real power absorbed} = 29.04 \text{ kW}$$

(c) Total complex supplied by the source is

$$S = 29.04 - j58.08 \text{ kVA}$$